

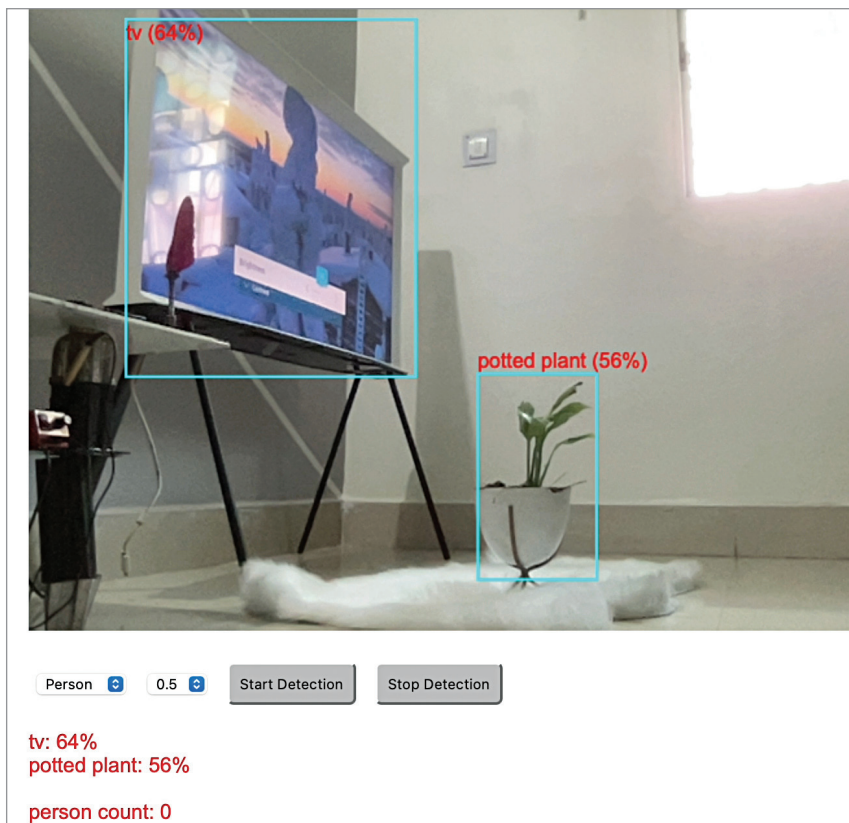


Fig. 6: Real-time detection of a television and a potted plant

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integrates well with the ESP32 platform. The IndusBoard Coin's 4MB PSRAM is critical for buffering image data, as the internal 320kB

SRAM is insufficient for real-time video processing. The PSRAM ensures continuous frame capture and reliable streaming without performance drops.

Software

The source code is written in Arduino IDE. With ESP32 board support installed, begin by including the necessary WebServer and Camera libraries. These enable the IndusBoard to operate as a local web server for live streaming. GPIO pin mappings are defined in the

initial section of the code, as shown in Fig. 4.

Subsequently, input the Wi-Fi network credentials—SSID and password—into the code to enable wireless connectivity. This allows the video stream to be accessed over a local network. Fig. 5 shows where to input these credentials within the code.

Testing

Once assembled as per Fig. 3 and flashed using the Arduino IDE, the device is powered and monitored via serial monitor to obtain the IP address. This address is then entered in a web browser to access the live stream.

During testing, the camera successfully detected a toy vehicle, a pet, and a human subject. Bounding boxes appeared over the detected objects. Better lighting improved detection accuracy, and adjusting the confidence threshold helped reduce false positives. Occasional streaming lag was observed due to Wi-Fi bandwidth limitations, but overall performance confirmed the feasibility of real-time object detection on a device of this scale. Fig. 6 shows the camera identifying a television and a potted plant simultaneously. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-make-your-own-smallest-spy-camera-with-ai>

Ashwini Kumar Sinha, an IoT and AI enthusiast, is Tech Journalist at EFY.

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